Acceleration

Piezoelectric acceleration sensors

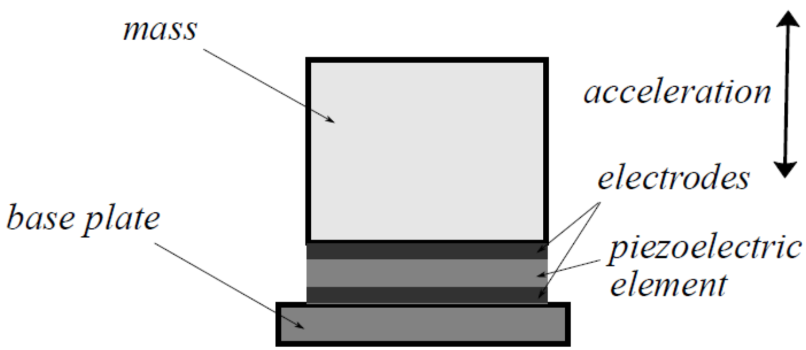


FIGURE 9.2: Principle of piezoelectric acceleration sensor

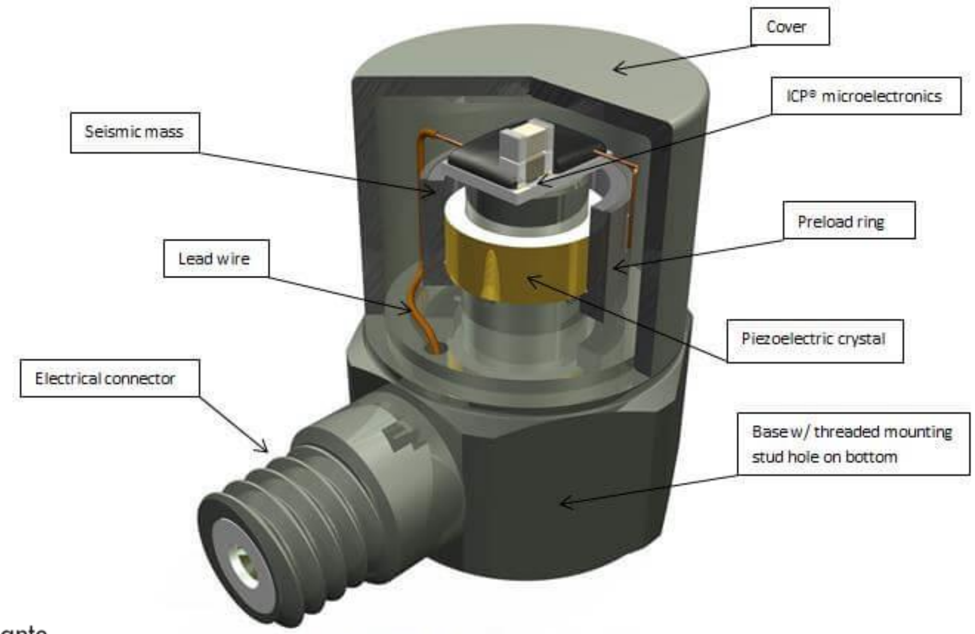
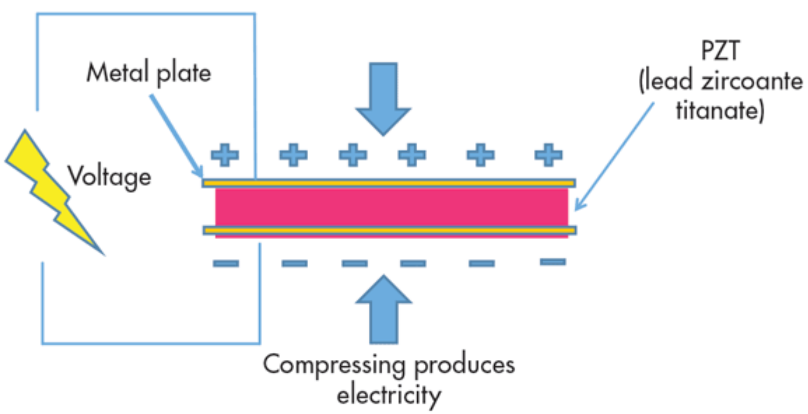


Figure 1: Typical ICP® Accelerometer



Piezoresistive acceleration sensors

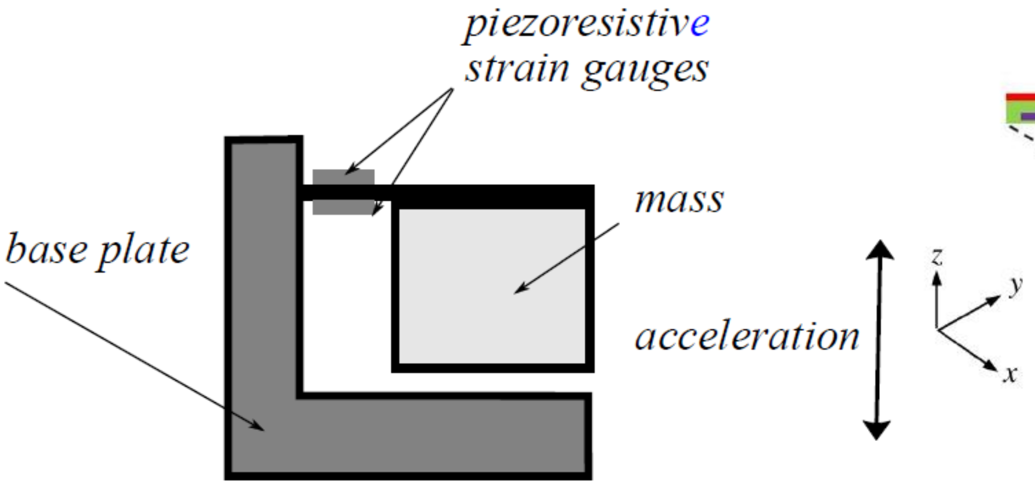
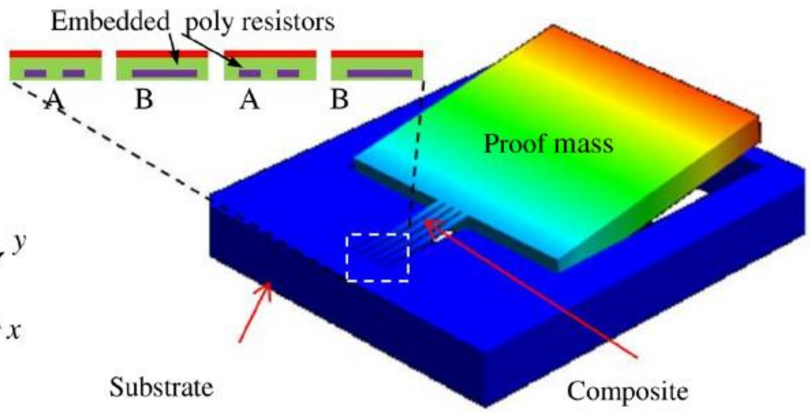
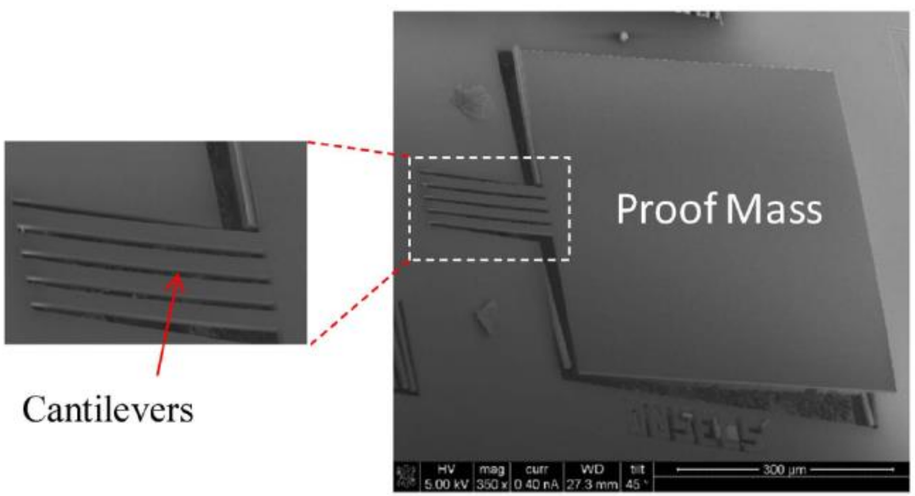


FIGURE 9.5: Piezoresistive acceleration sensor





Acceleration sensors with measured displacement

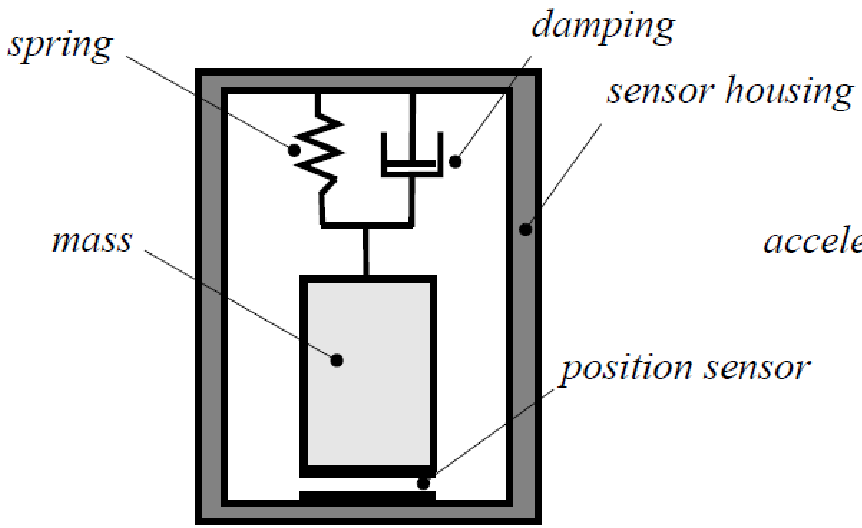


FIGURE 9.8: General principle of acceleration sensors with measured displacement

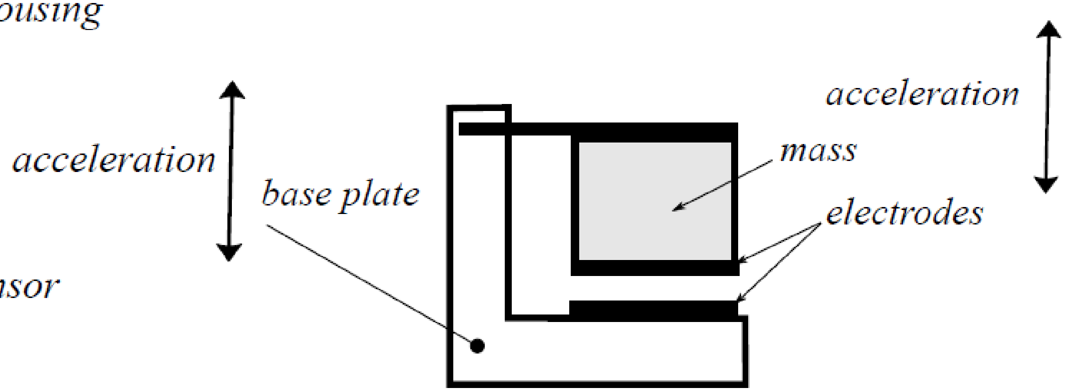
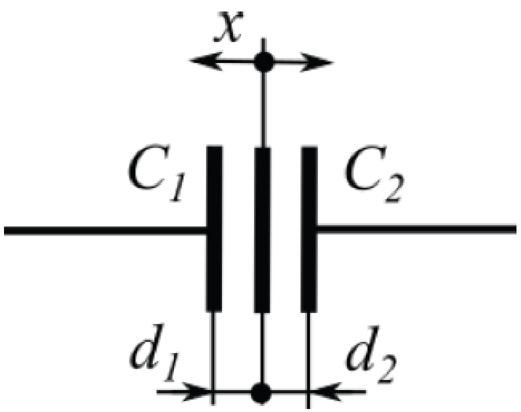
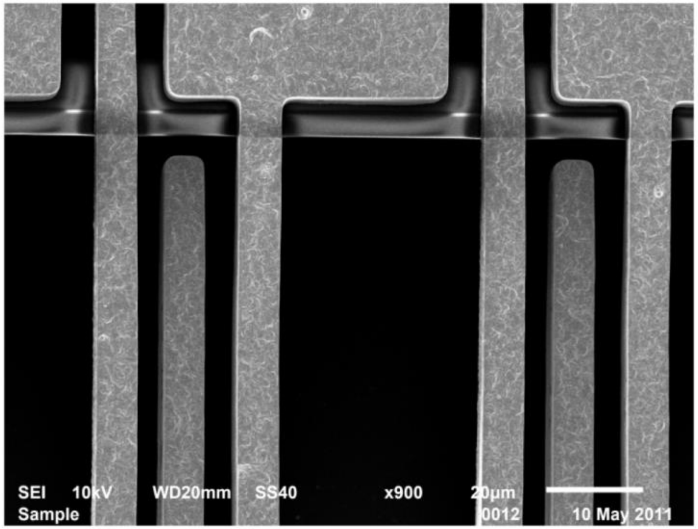
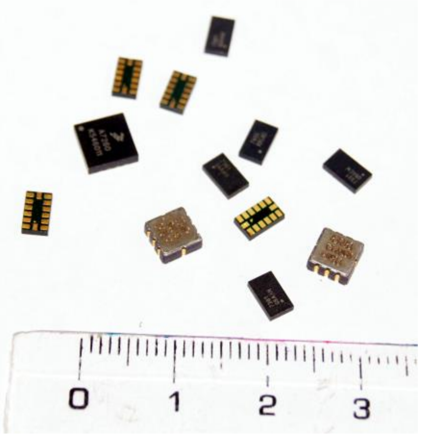


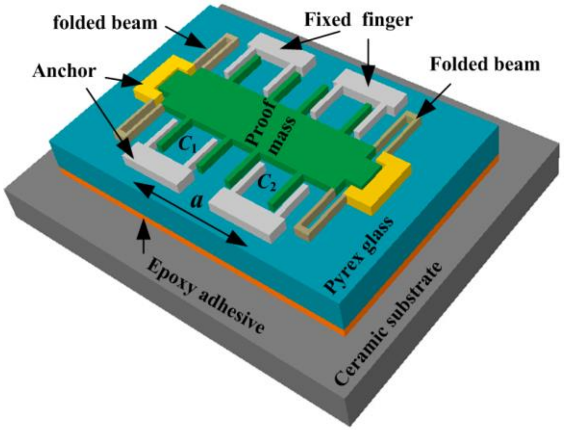
FIGURE 9.9: Capacitive accelerometer

Acceleration sensors with measured displacement

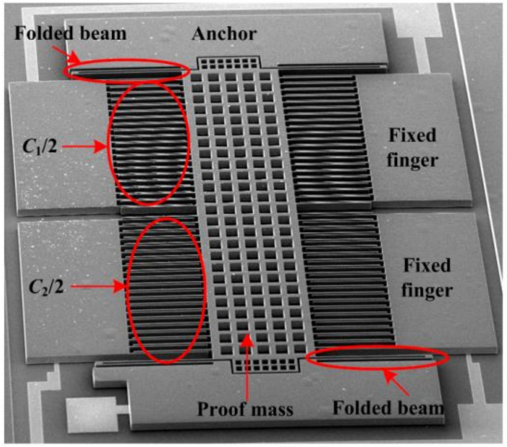






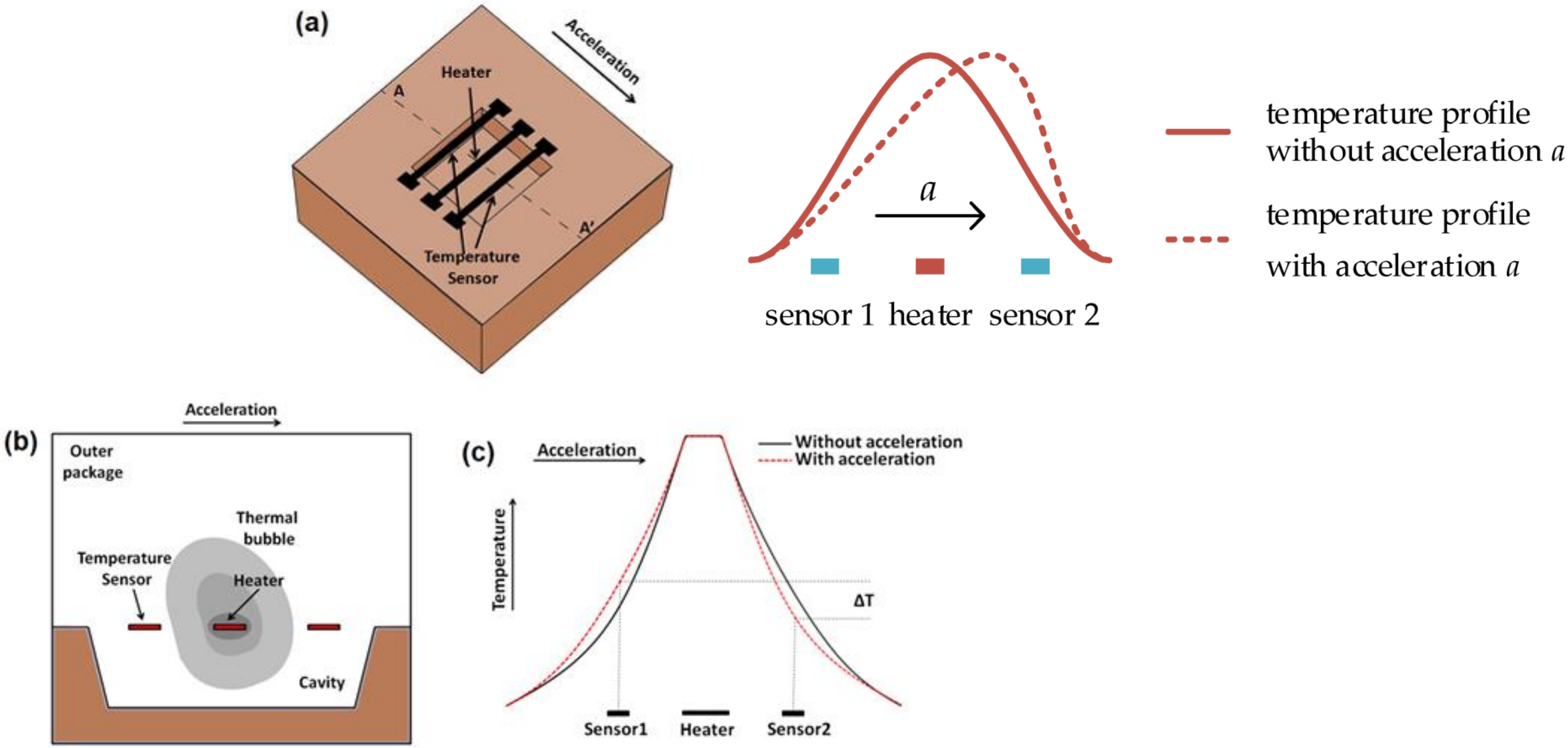


(a)



(b)

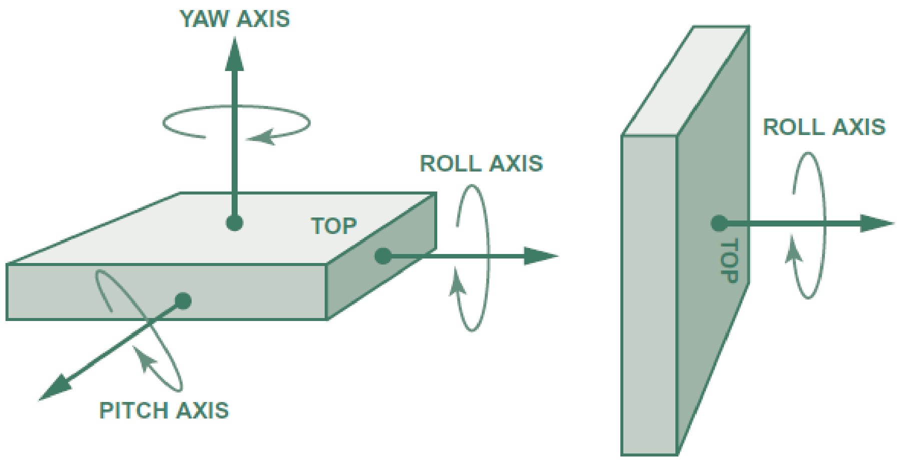
Thermal gas gyroscope



- Mukherjee, Rahul, et al. "A review of micromachined thermal accelerometers." *Journal of Micromechanics and Microengineering* 27.12 (2017): 123002.
- Liu S, Zhu R. Micromachined Fluid Inertial Sensors. *Sensors*. 2017; 17(2):367. https://doi.org/10.3390/s17020367

Gyroscope

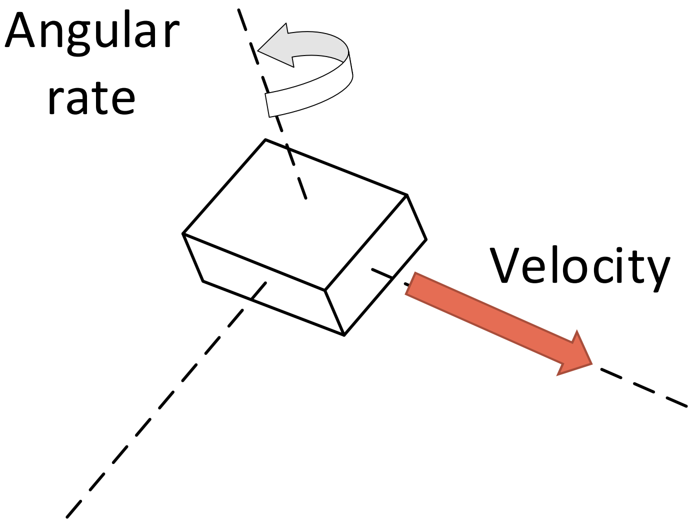
- Gyroscope is a type of inertial sensor that measures the angular rate

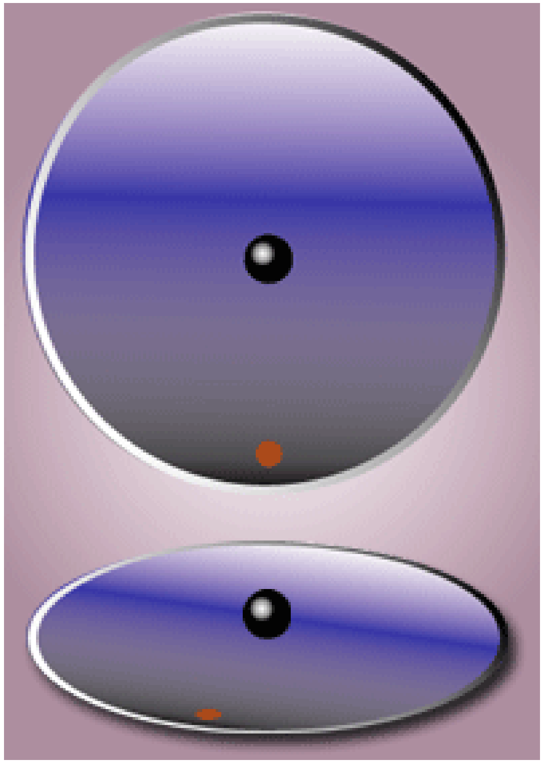


- Applications:
  - Measure how quickly an object turns
  - Integrate the angular rate over time to determine angular position

Gyroscope

- Conventional Vibratory Gyroscope:
  - Measure angular rate by means of Coriolis effects



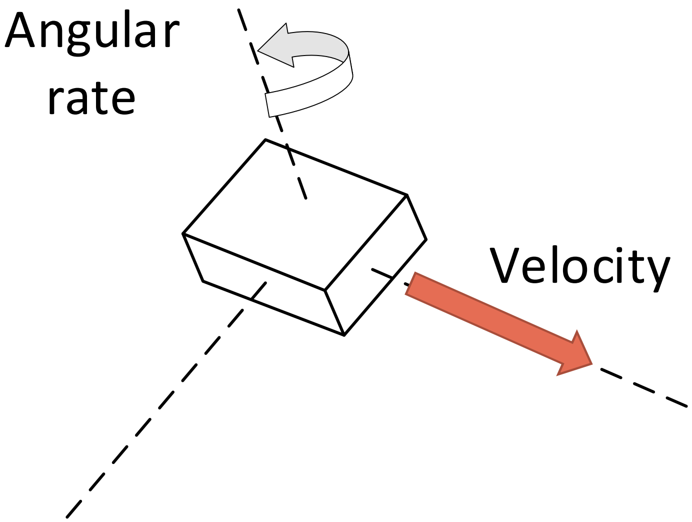


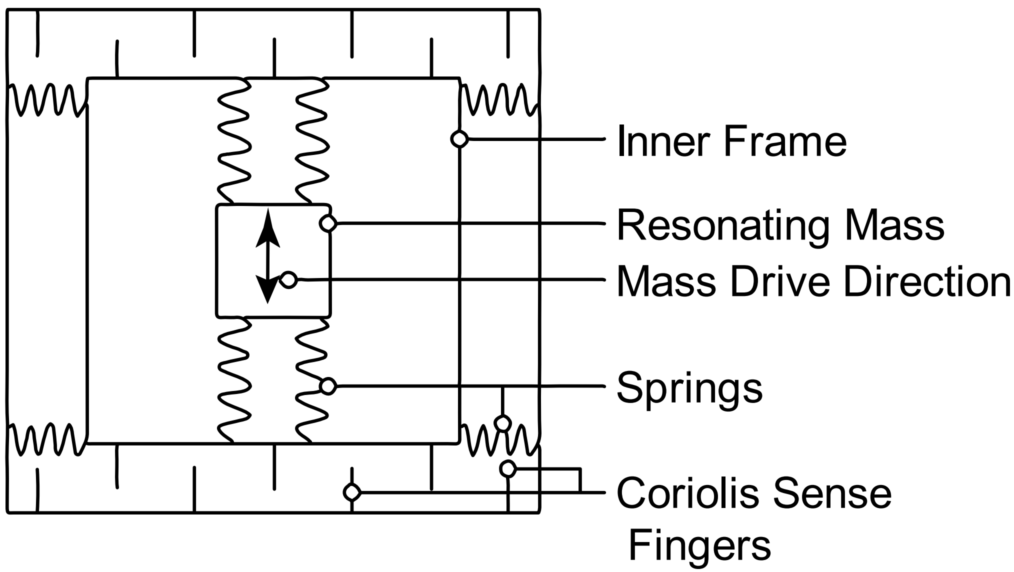
$F_C = -2m(\Omega \times v)$

- m: mass
- Ω: angular velocity
- v: tangential velocity

Gyroscope

- Conventional Vibratory Gyroscope:
  - Measure angular rate by means of Coriolis effects





$$F_C = -2m(\Omega \times v)$$

- m: mass
- Ω: angular velocity
- v: tangential velocity